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1  #!/usr/bin/env python3
2  # -*- coding: utf-8 -*-
3  """
4  Example 3.3
5  Early Stopping
6  Machine Learning for Engineering Problem Solving
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8  """
9
10 import IPython as IP
11 IP.get_ipython().run_line_magic('reset', '-sf')
12
13 import numpy as np
14 import matplotlib.pyplot as plt
15 import sklearn as sk
16
17 plt.close('all')
18
19 %% build the data sets
20
21 # use 6 to help give a smooth curve that makes the case for early stopping
22 np.random.seed(6)
23
24 m = 20
25 X = 6 * np.random.rand(m, 1) - 3
26 Y = 0.5 * X**2 + X + 2 + np.random.randn(m, 1)
27
28 # plot the data
29 plt.figure()
30 plt.grid(True)
31 plt.scatter(X,Y,color='gray')
32 plt.xlabel('x')
33 plt.ylabel('y')
34
35 X_model = np.linspace(-3,3,num=1000)
36 X_model = np.expand_dims(X_model,axis=1)
37
38 X_train, X_val, y_train, y_val = sk.model_selection.train_test_split(X, Y, test_size=0.2)
39
40 %% perform early stopping
41
42
43 # prepare the data
44 poly_scaler = sk.pipeline.Pipeline([("poly_features", sk.preprocessing.PolynomialFeatures
45 (
46     degree=90, include_bias=False)), ("std_scaler", sk.preprocessing.StandardScaler())])
47 X_train_poly_scaled = poly_scaler.fit_transform(X_train)
48 X_val_poly_scaled = poly_scaler.fit_transform(X_val)
49
50 # set up the model, not that by setting max_iter=1 it will only train one epoch
51 model = sk.linear_model.SGDRegressor(max_iter=1, tol=0, learning_rate="constant"
52 ,eta0=0.0005,penalty=None,warm_start=True)
53
54 # Train the model in a loop to build the data set to investigate the benefit of early
55 # stopping
56 val_errors = []
57 train_errors = []
58 for epoch in range(1000):
59     model.fit(X_train_poly_scaled, y_train.ravel()) # continues where it left off
60     y_val_predict = model.predict(X_val_poly_scaled) # Predict the target values
61     y_train_predict = model.predict(X_train_poly_scaled) # Predict the target values
62     val_error = sk.metrics.mean_squared_error(y_val, y_val_predict) # Calculate error
63     train_error = sk.metrics.mean_squared_error(y_train.ravel(), y_train_predict) #
64     Calculate error
65     val_errors.append(val_error)
66     train_errors.append(train_error)

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65
66 # plot the early learning curves, you may have to plot this a few times to get
67 # a set of curves that shows strong results
68 plt.figure()
69 plt.grid(True)
70 plt.plot(val_errors, label='validation data')
71 plt.plot(train_errors, '--', label='training data')
72 plt.ylabel('RMSE')
73 plt.xlabel('epoch')
74 plt.legend()
75
76
77
78
79
80
```